J. Geomag. Geoelectr., **38**, 325-334, 1986

Differential Rotation of Northeastern Part of Southwest Japan: Paleomagnetism of Early to Late Miocene Rocks from Yatsuo Area in Chubu District

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(Received January 14, 1986)

A paleomagnetic study has been carried out on the Miocene volcanic and sedimentary rocks from the Yatsuo area, the northern part of the Chubu district. Stable primary magnetizations are recognized from 8 sites among 36 sites ranging from 17 to 7 Ma in age. Eight sites are grouped into two by their declinations; the lower 4 sites show easterly deflected declinations (mean direction: $D=12.6°$, $I=47.1°$, $\alpha_{95}=7.0°$) and the upper 4 sites show no significant deflection from the axial geocentric dipole field (mean direction: $D=-2.5°$, $I=55.6°$, $\alpha_{95}=6.6°$). This result suggests that the Yatsuo area was subjected to a clockwise rotation about 13° between 15 and 12 Ma, possibly synchronous with the rotation of Southwest Japan. The smaller rotation angle than that estimated for Southwest Japan (47°) suggests the differential rotation between the block which contains the Yatsuo area and the main part of Southwest Japan.

1. Introduction

OTOFUJI *et al.* (1985) asserted that Southwest Japan had rotated clockwise as much as 47° synchronously with the opening of the Japan Sea in Middle Miocene. They situated the eastern boundary of the rotated block near Toyama in the Chubu district (Fig. 1). However, paleomagnetic data from this district are not enough to fix the eastern boundary of rotated Southwest Japan.

Thick Neogene sequences composed of volcanic and sedimentary rocks are exposed in the northern coastal area of the Chubu district along the Japan Sea. The Neogene rocks in the Yatsuo area (Fig. 2) are well-exposed, and some detailed biostratigraphic works have been performed (HAYAKAWA, 1983; 1985). Results of a paleomagnetic study on the Neogene rocks exposed in this area are described in this report.

2. Geologic Setting and Sampling

The Yatsuo area is underlain by lower Miocene volcanics (so-called Green Tuff) and marine sedimentary rocks from Miocene to Pleistocene in age. These rocks generally dip 10° - 40° to the north with neither apparent tectonic disturbance nor slumping structure.

The Neogene System in the Yatsuo area is divided into two groups, namely, the Yatsuo Group and the Tonami Group in ascending order (HAYAKAWA, 1985). The

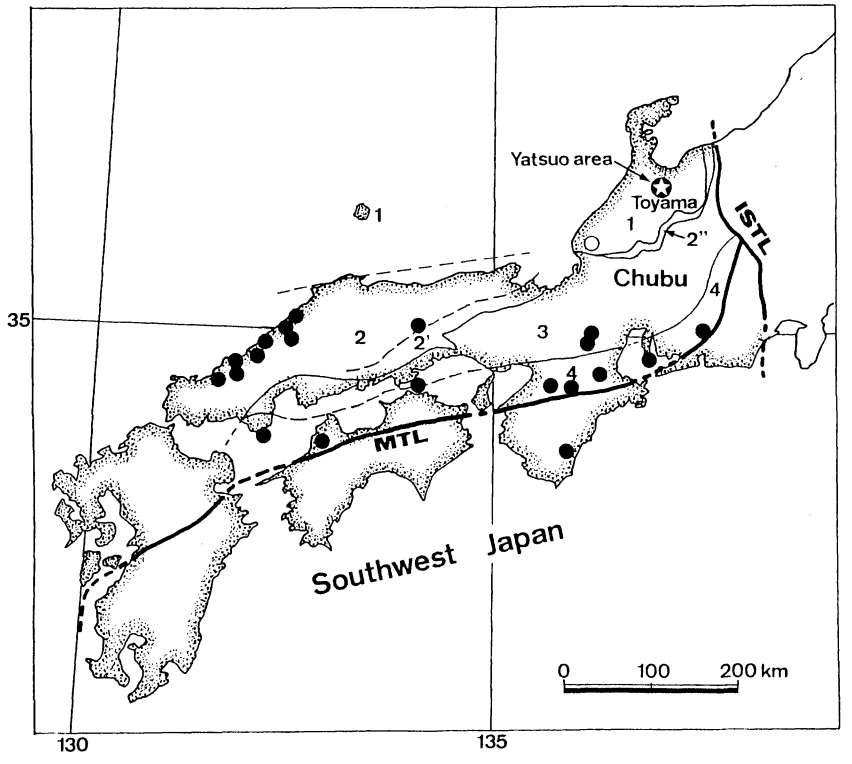


Fig. 1. Summary of geotectonic provinces in Southwest Japan and sites of Tertiary paleomagnetic data. 1: Hida belt. 2: Sangun belt. 2': Maizuru structural belt. 2": Hida marginal belt. 3: Mino-Tanba belt. 4: Ryoke belt. MTL: Median tectonic line. ISTL: Itoigawa-Shizuoka tectonic line. Solid circle: OTOFUJI *et al.* (1985). Open circle: NAKAJIMA and HIROOKA (1986).

Yatsuo Group is unconformably underlain by the Hida gneisses and the early Mesozoic Funatsu granites. The Yatsuo Group, from the base upward, consists of coarse clastic sediments, mafic to felsic volcanic rocks and a thick series of marine sediments. This group is lithologically subdivided into the following five units; Nirehara, Iwaine, Iozen, Kurosedani and Higashibessho Formations in ascending order. The lithofacies of these units are schematically shown in Fig. 3. The Tonami Group is unconformably underlain by the Yatsuo Group. The Tonami Group mainly consists of coarse clastic sediments. This group is lithologically subdivided into three units, which are the Tenguyama, Otogawa, and Mita Formations in ascending order (Fig. 3).

Biostratigraphic data obtained from the Yatsuo Group were compiled by HAYAKAWA and TAKEMURA (in preparation). Planktonic foraminifera assemblages yielded from the Kurosedani Formation suggest that this formation is assigned to the Zone N 8 (15.5–16.5 Ma) of BLOW (1969). Diatom assemblages from the Higashibessho Formation suggest that this formation is assigned to the *Denticulopsis lauta* Zone (14.85–15.70 Ma) of KOIZUMI (1985). Three radiometric ages have been reported from the Yatsuo Group (Fig. 3). A K–Ar age was determined to be 16.4 Ma for the

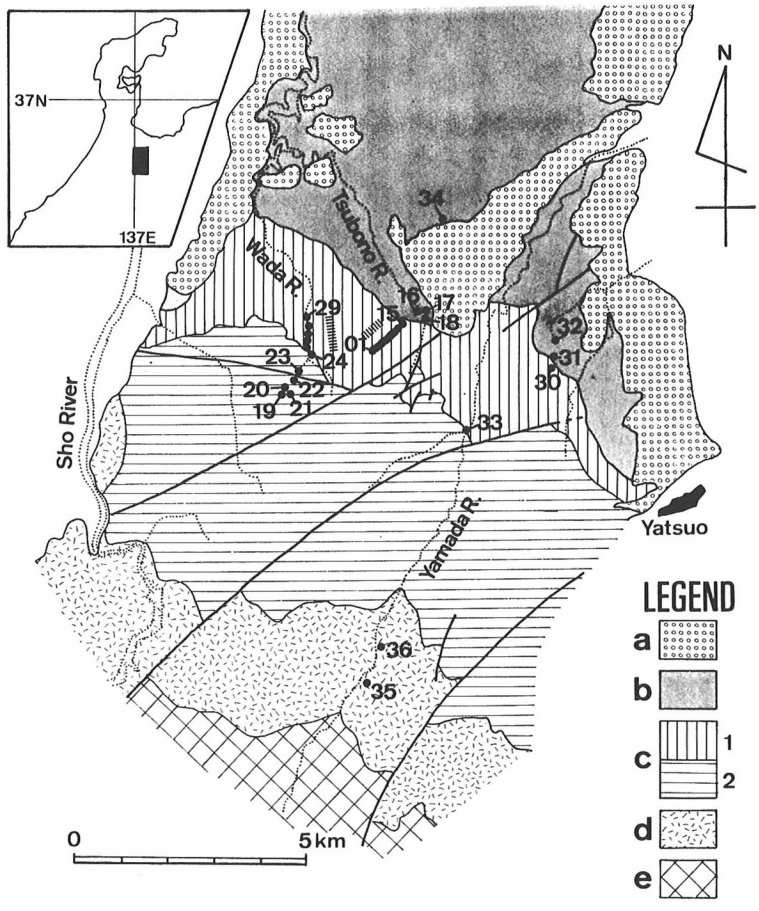


Fig. 2. Geologic map simplified from HAYAKAWA (1985) and the localities of paleomagnetic sampling sites. a: Quaternary deposits. b: Tonami Group. c: upper part of the Yatsuo Group (1: Higashibessho Formation. 2: Kurosedani Formation). d: lower part of the Yatsuo Group. e: basement rocks.

andesite lava from the Iwaine Formation (SHIBATA, 1973). The fission-track age of the Yamadanaka tuff (uppermost part of the Kurosedani Formation) is 16.0 Ma (HAYAKAWA, 1983). The Yamadanaka tuff was also dated to be 15.0 Ma by fission-track method (NAKAJIMA and MIZUSHIMA, 1984). These radiometric ages are concordant with the biostratigraphic age estimation.

HAYAKAWA (1985) reported two biostratigraphic data from the Tonami Group (Fig. 3). The Mita Formation yields characteristic molluscan fossils (Omma Fauna) that is of late Pliocene to early Pleistocene age. A diatom assemblage found from the middle part of the Otogawa Formation suggests that this horizon is correlatable to the *Thalassionema schraderi* Zone (7.30–9.20 Ma) of KOIZUMI (1985). There is no report on age-diagnostic fossils for the Tenguyama Formation. One fission-track age is determined to be 12.3 Ma for a pumice tuff from the lower part of the formation (HAYAKAWA, 1983).

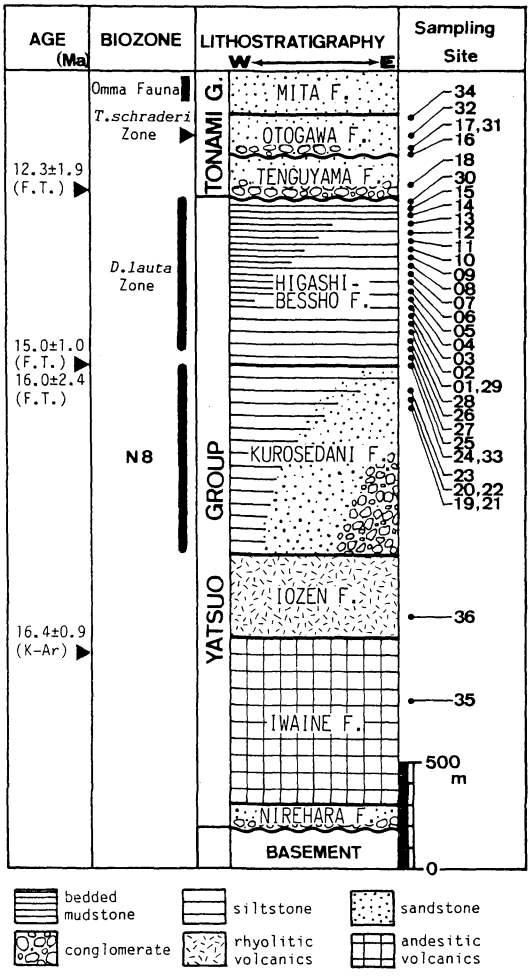


Fig. 3. Summary of lithostratigraphy (HAYAKAWA, 1985), biostratigraphy (HAYAKAWA, 1985; HAYAKAWA and TAKEMURA, in preparation), radiometric age (SHIBATA, 1973; HAYAKAWA, 1983; NAKAJIMA and MIZUSHIMA, 1984), and the horizons of paleomagnetic sampling sites.

Geologic mapping at 1:25,000 scale (HAYAKAWA, 1985) provides regional structural and stratigraphic control for the sampling sites. Samples were collected at 36 sites as shown in Figs. 2 and 3. In this study, a site was defined as a bed (less than 2 m thick) which was exposed in a continuous outcrop wider than 5 m. Each site comprises 4 to 10 hand samples, which were independently oriented using a magnetic compass.

3. Laboratory Procedure and Result

Cylindrical specimens of 25 mm in diameter and 22 mm in length were cut from each hand sample. Either a Schonstedt SSM-1A spinner magnetometer or a ScT C-112 cryogenic magnetometer were used to measure the remanent magnetization.

Two specimens were selected from each site, and their magnetic stability was tested by means of progressive demagnetization with thermal and alternating field methods. Alternating field demagnetization was carried out stepwise up to 160 mT with a two-axis tumbler contained in a three-layered mu-metal shield. Progressive thermal demagnetization was carried out in more than 12 steps up to 700°C. Specimens were heated in air using a non-inductively wound electric furnace. The stray field during the cooling cycle was reduced to less than 10 nT by a three-layered mu-metal envelope.

Using the vector-demagnetization diagrams for progressive thermal demagnetization of pilot specimens, the following sites were excluded from the routine demagnetization procedure: (1) Progressive change of the vector end-points did not show a straight trend. Secondary magnetizations were not removed effectively for this case. (2) Stable magnetic direction coincided with the present geomagnetic field direction before the tilt correction. Such component might be of the secondary origin. (3) The stable component did not converge to the original point of the diagram. In such site, secondary magnetizations prevented the determination of the high-temperature components.

Eight sites among 36 survived the criteria. These sites were assumed to have primary magnetizations. For the 8 sites, progressive alternating field demagnetization was also applied. In 5 sites of the eight, the primary components detected by thermal demagnetization were also recognized by means of alternating field demagnetization (Fig. 4-a). For these 5 sites, alternating field demagnetization was adopted for cleaning of the rest of the specimens. For the other 3 sites, however, alternating field demagnetization was not effective to separate the primary component (Fig. 4-b). Thermal demagnetization, therefore, was adopted for the cleaning of the rest of the specimens.

The partial demagnetization was made for 8 to 12 specimens per each site at 3 to 5 steps of demagnetization levels. An optimum demagnetization level was decided as to give the minimum within-site scatter of direction (minimum α_{95}). The α_{95} values were less than 10° at the optimum demagnetization levels for every site. The site-mean magnetic directions (after tilt correction) determined at optimum demagnetization levels are shown in Table 1.

The paleomagnetic declinations are shown stratigraphically in Fig. 5. As clearly shown in the figure, the paleomagnetic directions of lower sites have eastward deflected declinations, while upper sites show no significant deflection. The paleomagnetic directions, therefore, are grouped into two categories by their declinations: the lower 4 sites from the Yatsuo Group, and the upper 4 sites from the Tonami Group. The formation-means are $D=12.6°$, $I=47.1°$, and $\alpha_{95}=7.0°$ (the lower sites), and $D=-2.5°$, $I=55.6°$, and $\alpha_{95}=6.6°$ (the upper sites). A statistical test for discrimination (MCFADDEN and LOWES, 1981) shows the distinction of the two populations. The null hypothesis that the lower and the upper sites share a common true mean direction is rejected with 95% confidence.

4. Interpretation

The periods covered by both of the lower and the upper sites are sufficiently long to average out the effect of geomagnetic secular variation, so that the formation-

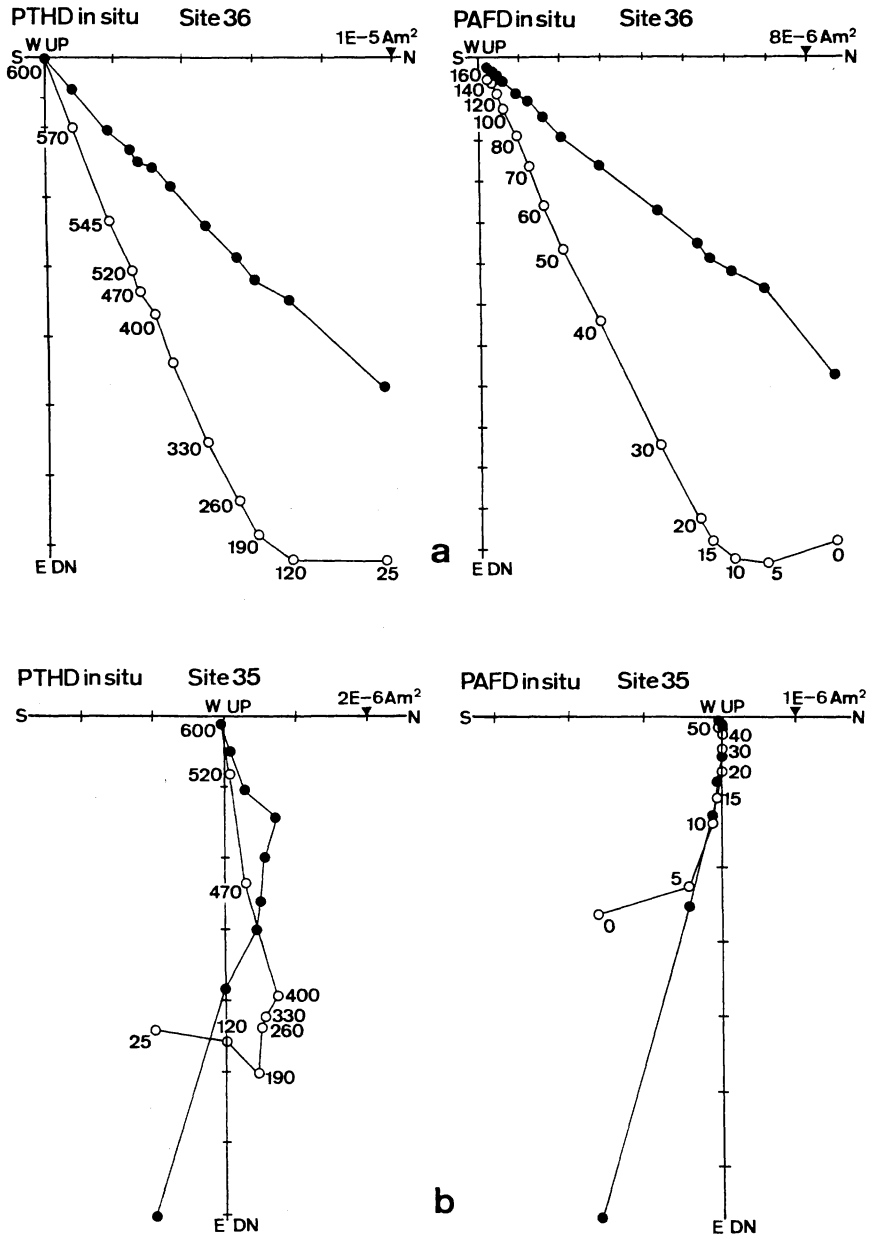


Fig. 4. Typical vector-demagnetization diagrams of progressive thermal (PTHD) and progressive alternating field (PAFD) demagnetization of specimens from Site 36 (a) and Site 35 (b). Solid and open circles are projections of vector end-points on horizontal and N–S vertical planes, respectively. Numbers attached to symbols are temperature in °C and peak alternating field in mT. (a) Magnetization is regarded as primary component except for the soft secondary component removed up to 190°C or 20 mT. (b) Secondary component cannot be removed thoroughly by alternating field demagnetization, while thermal demagnetization presents stable primary component above 400°C.

Table 1. Tilt-corrected site mean directions from the Miocene Series in the Yatsuo area.

Site	DEMAG.	Dec.	Inc.	N	α_{95}	k	Polarity	PHI	LMD
05	20 mT	-172.5	-51.7	12	3.2	190.1	R	82.5	-100.5
16	10 mT	6.9	58.0	10	2.0	572.0	N	84.2	-155.6
18	10 mT	-0.1	53.4	8	4.6	145.1	N	87.3	-41.1
31	20 mT	-10.4	59.4	11	7.4	39.2	N	81.1	74.2
32	120 °C	-6.4	50.8	8	3.2	294.4	N	82.7	5.1
33	190 °C	-174.0	-45.7	12	4.9	80.1	R	79.3	-72.9
35	330 °C	17.9	41.7	9	4.3	143.1	N	70.2	-98.8
36	20 mT	18.3	48.7	8	2.0	790.8	N	73.2	-113.9

DEMAG: demagnetization level in °C or mT. Dec., Inc.: mean declination and inclination in degree. N: number of specimens. α_{95}: radius of the 95% confidence circle in degree. k: Fisher's precision parameter. PHI,LMD: latitude and longitude of virtual geomagnetic pole position (north-seeking pole) in degree.

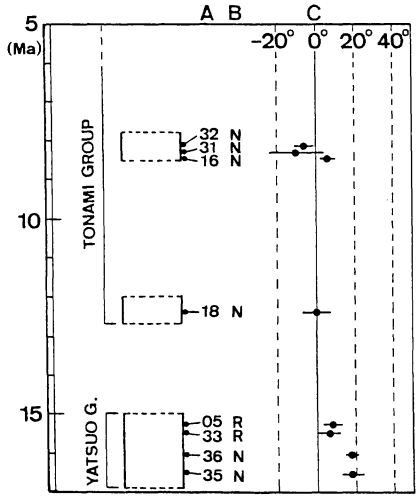


Fig. 5. Stratigraphic distribution of paleomagnetic data. A: site number. B: polarity. C: declination and its uncertainty [arcsin ($\sin\alpha_{95}/\cos I$), (KELLOG and REYNOLDS, 1978)]

means of the lower and the upper sites can be regarded as characteristic directions of these rock units: the most part of the Yatsuo Group, and the lower part of the Tonami Group (Fig. 5).

The difference of characteristic directions is not attributed to the effect of fault motion or folding inside of the surveyed area. There is no serious regional deformation of the Neogene rocks in the Yatsuo area. The effect of polar wander also cannot explain the difference of characteristic directions as the late Tertiary paleomagnetic poles of Eurasia (IRVING and IRVING, 1982) almost coincide with the present geographic pole. The difference must be attributed to a rotation of a block which contains the Yatsuo area.

The rotation angles (BECK, 1980) are 12.6° for the lower sites, and −2.5° for the upper sites. In this study, expected declination is calculated from the axial dipole. This

result suggests that the Yatsuo area rotated clockwise (about 13°) around vertical axis. It seems that the inclination values of the lower and the upper sites do not suggest significant north–south translation, considering the effect of northward dipole offset in Tertiary (WILSON, 1971).

As shown in Section 2, the upper part of the Yatsuo Group is correlated to the Zone N 8 (15.5–16.5 Ma) and the *Denticulopsis lauta* Zone (14.85–15.70 Ma). It is suggested, therefore, the clockwise rotation of the Yatsuo area occurred after about 15 Ma. This rotation finished before the deposition of the lower part of the Tonami Group. Chronological data obtained from the Tonami Group, previously shown in Section 2, suggest that the rotation ceased before about 12 Ma.

Bio- and chrono-stratigraphic correlation (Fig. 6) suggests that the Tonami Group was formed after the period of the rotational event of Southwest Japan. It is also shown that the most of the Yatsuo Group is correlative to the Miocene Series in both the Kumano area and Ichishi area, which are included in the data summarized by OTOFUJI *et al.* (1985). Mean paleomagnetic directions of these areas show an eastward deflection of more than 40°. This result suggests that the rotation of the Yatsuo area took place during the period similar to that of Southwest Japan. It is most likely that the rotational movement of the Yatsuo area has a relation to the clockwise rotation of Southwest Japan.

Rotation angle of the Yatsuo area (about 13°) is much smaller than that of Southwest Japan (about 47°; OTOFUJI *et al.*, 1985). From this difference (34°), it is

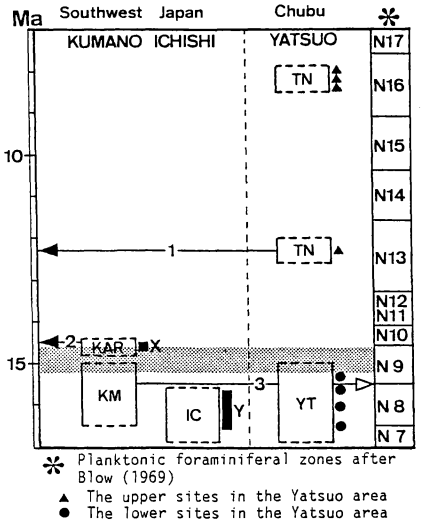


Fig. 6. Bio- and chrono-stratigraphic correlation between the Yatsuo area and Southwest Japan. Numbers adjacent to arrows indicate data sources for the correlation (1: HAYAKAWA, 1983. 2, 3: TSUCHI *et al.*, 1981). Shaded zone indicates the duration of rotational movement of Southwest Japan (OTOFUJI *et al.*, 1985). *X* and *Y* show time distributions of paleomagnetic data obtained from the Kumano area (TAGAMI, 1982) and the Ichishi area (HAYASHIDA and ITO, 1984), respectively. Mean declinations and their uncertainties are 57.5° ± 33.0° for Kumano area, and 45.1° ± 18.2° for Ichishi area. KAR: Kumano Acidic Rocks. KM: Kumano Group. IC: Ichishi Group. TN: Tonami Group. YT: Yatsuo Group.

proposed that the differential rotation occurred between the block which contains the Yatsuo area and the main part of Southwest Japan.

In the northern part of the Chubu district, the other paleomagnetic data from Tertiary rocks is reported after careful demagnetizations and tilt corrections. NAKAJIMA and HIROOKA (this issue) reported paleomagnetic directions from Miocene rocks in the Niu Mountains, the westernmost part of the Hida belt (Fig. 1). They concluded that the Niu Mountains were subjected to a clockwise rotation through 42°, which occurred between 16 and 15 Ma. The Niu Mountains are, therefore, presumed to be rotated as a part of Southwest Japan. The rotational movement of the Niu Mountains seem almost synchronous with that of the Yatsuo area, but the amount of the rotation angles are quite different. This difference may be caused by the deformation of the Hida belt, basement rocks of the area, in Middle Miocene. HIROOKA *et al.* (1985) advocated the deformation of the Hida belt on the basis of paleomagnetic data of Menozoic rocks, such as the Tetori Group of late Jurassic to Cretaceous age. Those paleomagnetic directions suggest that the Hida belt deformed some time later than Cretaceous to form the northward curvature of the eastern part of the Hida belt. Paleomagnetic data obtained from the Yatsuo area suggest that the deformation of the Hida belt occurred in Middle Miocene.

5. Conclusion

(1) Paleomagnetic directions were determined for 8 sites in the Yatsuo area. The lower 4 sites from the Yatsuo Group range from 17 to 15 Ma in age. The formation-mean direction shows easterly deflected declination ($D=12.6°$, $I=47.1°$, $\alpha_{95}=7.0°$). The upper 4 sites from the Tonami Group range from 12 to 7 Ma in age. The formation-mean direction is almost parallel to the axial geocentric dipole field ($D=-2.5°$, $I=55.6°$, $\alpha_{95}=6.6°$).

(2) The Yatsuo area has undergone about 13° clockwise rotation. This rotational movement started after about 15 Ma, and ceased before about 12 Ma. The amount of the rotation of the studied area is much smaller than that of the main part of Southwest Japan (47°). This discrepancy is attributed to a differential rotation between the block which contains the Yatsuo area and the main part of Southwest Japan since 15 Ma.

I am much indebted to Masayuki Torii for his helpful advice. I wish to thank Hideki Hayakawa for his support during the field work. Thanks are also due to Ikuo Katsura and two anonymous reviewers for providing important suggestions.

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